

VELTRAXX'26

SHAKTI
JF ENGINEERING AND TECHNO
JTE, New Delhi. • Affiliated to Anna University, C
AN AUTONOMOUS INSTITUTION

PARTNERS



इलेक्ट्रॉनिकी एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
ELECTRONICS AND
INFORMATION TECHNOLOGY
सत्यमेव जयते



**Chips to Startup
Programme**



**INSTITUTION'S
INNOVATION
COUNCIL**
(Ministry of Education Initiative)

BAJAJ INSTITUTE OF TECHNOLOGY, WARDHA

TEAM NAME: YEAGERIST

PROBELM STATEMENT NO: PS02

PROBLEM STATEMENT TITLE: Multi-Master AXI4-Lite Arbiter

PS Category-
Software/Hardware

VELTRAXX'26

PROBLEM STATEMENT

- Four SoC masters may request the same shared bus simultaneously.
- Uncontrolled contention can corrupt transfers or misroute responses.
- Slow peripherals create back-pressure and unpredictable latency.
- Fixed priority can starve lower-priority masters.

VELTRAXX'26

YOUR SOLUTION

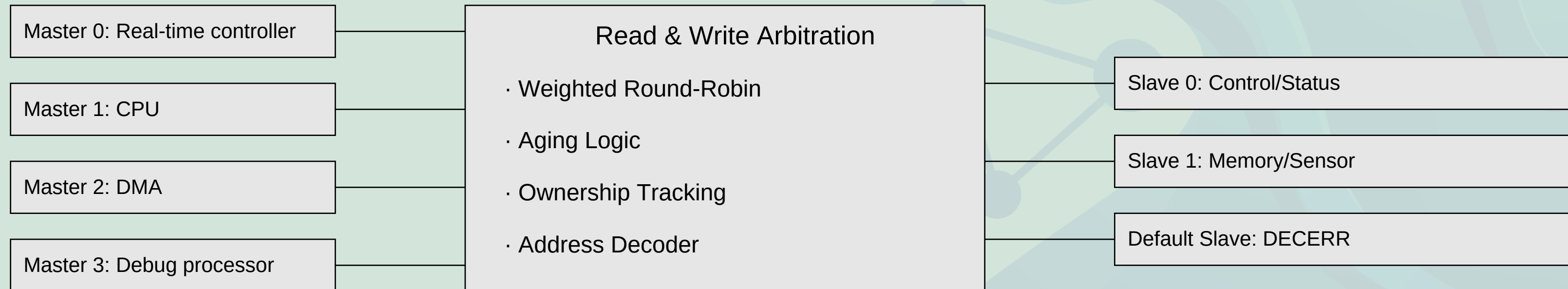
- Project name: AXI-GUARD
- A protocol-safe 4-master, 2-slave AXI4-Lite interconnect.
- Separate read/write arbiters allow concurrent traffic.
- Weighted round-robin provides controlled priority.
- Aging prevents indefinite starvation.
- Registered ownership keeps responses with the correct master.
- A default slave returns DECERR for unmapped addresses.



VELTRAXX'26

ARCHITECTURAL BLOCK DIAGRAM

○



VELTRAXX'26

PURPOSE OF BLOCKS

- Input: Captures independent AXI AW, W and AR requests.
- Processing: Arbitrates, decodes addresses and tracks ownership.
- Output: Returns B/R responses to the originating master.
- Architecture choice: Separate read/write paths preserve concurrency and ordering.
- Trade-off: Extra buffering adds area, but simplifies correctness and verification.

VELTRAXX'26

OUTPUTS AND RESULTS

- AXI VALID/READY behavior remains stable under back-pressure.
- AW and W channels work independently.
- Read and write traffic can progress concurrently.
- Aging prevents starvation under sustained contention.
- Invalid addresses return DECERR.
- Verification includes simulation, assertions, formal checks and synthesis.

VELTRAXX'26

RESEARCH AND REFERENCES

- ARM AMBA AXI4-Lite Protocol Specification, IHI 0022.
- Project repository: <https://github.com/chandrasekarc968-cpu/SparkAhead->
- Yosys synthesis documentation.
- OpenROAD physical-design flow documentation.